Predicting Customer Churn at Telecom

A Data-Driven Retention Strategy

Agenda

01 **Market Context**

Why churn is the clearest lever for growth

02 **Data & EDA**

What 100,000 customers reveal about churn

03 **Model Development**

Testing three approaches; selecting the best

04 **Results**

LightGBM test AUC-ROC 0.703

05 **Business Impact**

Profitable only when sharply targeted; offer cost is the key lever

06 **Recommendation**

Two-track campaign; phased rollout plan

Retention Costs a Fraction of Acquisition: Telecom's Clearest Lever for Growth

Market context

5-25x

more expensive to acquire a new customer than to retain an existing one

Telecom markets are saturated

Worldwide there are 112 mobile subscriptions per 100 people (ITU, 2024). With the market saturated, growth depends on keeping existing customers.

Industry churn runs 20-40% annually

Nguyen et al. (2022) put average annual telecom churn at 20-40%, so every year a large share of the base must be replaced just to stand still.

Sources: Gallo (2014), Harvard Business Review; ITU Facts and Figures (2024); Nguyen et al. (2022), PLOS ONE.

100,000 Customers, Two Linked Data Sources, One Clear Target

Data foundation

~100,000
customers

~100
behavioral & account features

1
target variable (churn)



Client.csv

Demographics
Account & plan type
Tenure & device info

joined on Customer ID



Record.csv

Usage & billing metrics
Call quality indicators
Churn target variable

Merged Dataset

~100K rows
~100 features
1 target

Half the Sample Churned: a Balanced Dataset for Learning Who Leaves

Target variable analysis

49.6%

churn rate

within the 31-60 day window

50.4% stayed

Definition

Churn = customer left within 31-60 days of the observation date. Classes are nearly balanced (49.6% / 50.4%).

Why AUC-ROC?

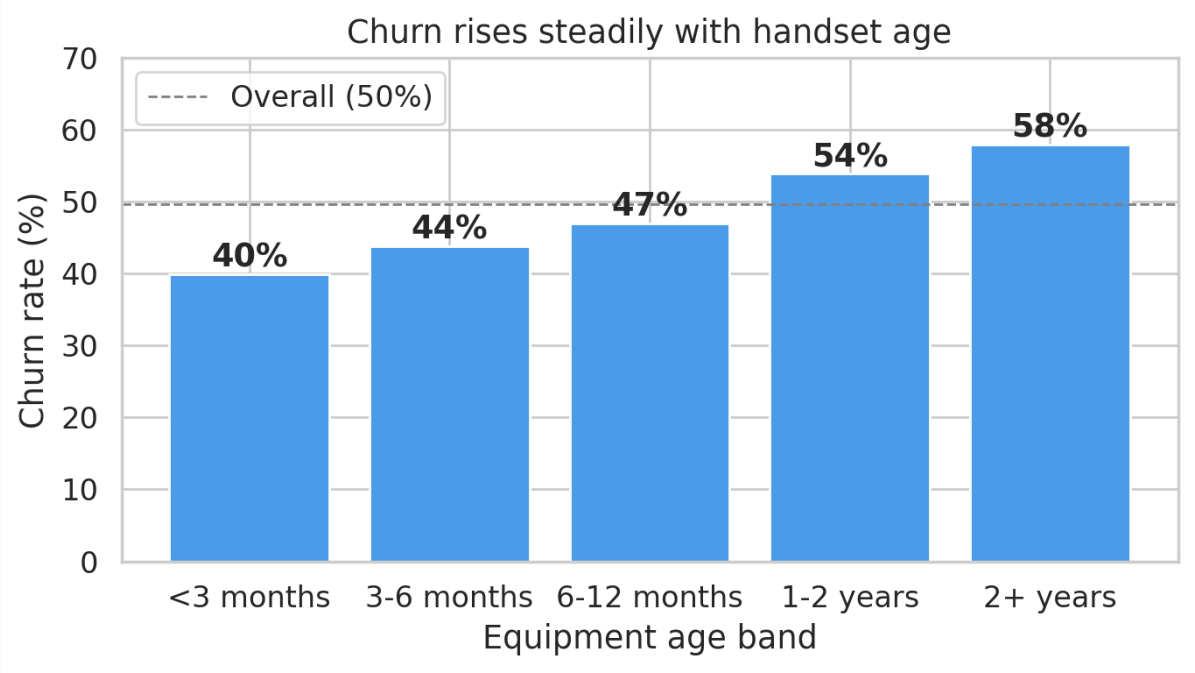
AUC-ROC is the primary metric: the campaign ranks customers by risk, and AUC measures ranking quality across all decision thresholds.

Read with care

~50% churn in two months is far above real-world rates (20-40% a year), so this is likely a balanced sample. Impact figures are flagged accordingly.

Churn Rises Steadily With Handset Age: 40% to 58%

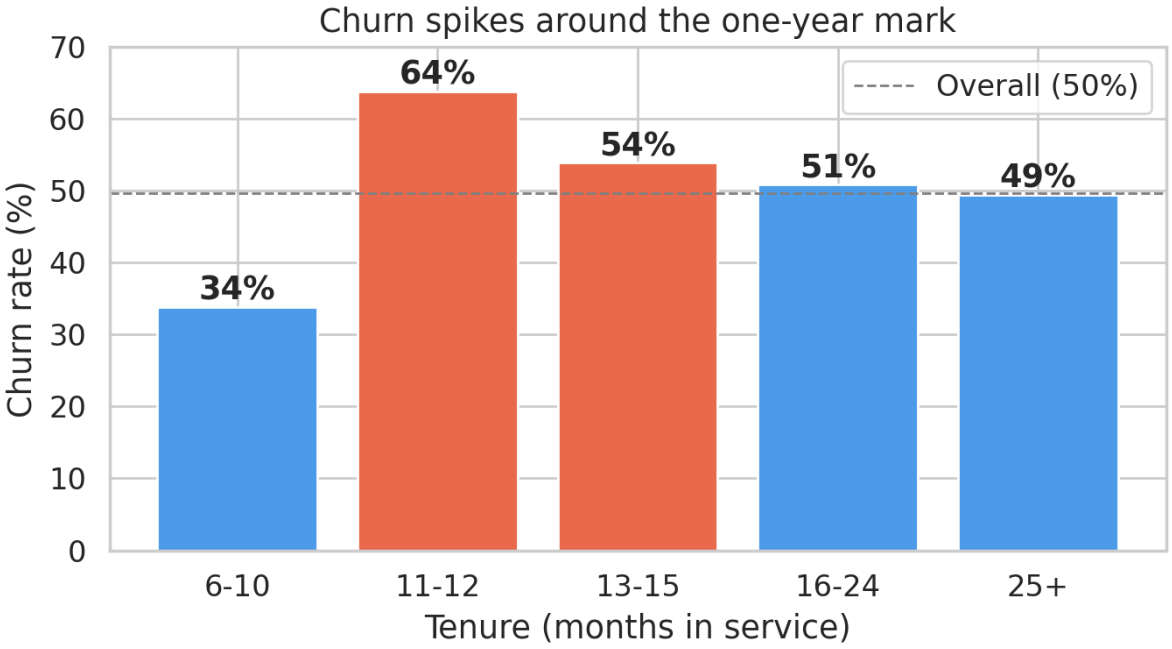
EDA: churn rate by equipment age band



The older the handset, the higher the churn rate: a clear, monotonic effect and a direct lever: an upgrade offer.

Churn Nearly Doubles at the One-Year Mark: 34% to 64%

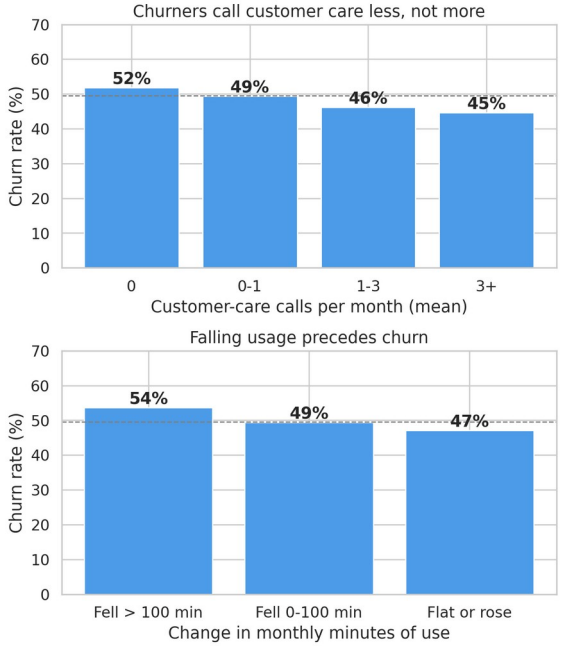
EDA: churn rate by tenure (months in service)



Consistent with 12-month contracts expiring: customers become free to leave. Contact them before month 11.

No Single Factor Predicts Churn, and Churners Rarely Complain First

EDA to model bridge



Churners leave quietly

Customers who never call customer care churn more (52%) than frequent callers (45%), so waiting for complaints misses them.

Falling usage is a warning sign

Minutes down 100+ vs the prior 3 months: 54% churn, vs 47% for stable users. Real, but modest.

Combination is the answer

Combining ~100 weak signals through gradient boosting produces a ranking no simple business rule could replicate.

Predicting Churn 31-60 Days Out Gives the Business a Real Window to Act

Business Question

Which customers are likely to leave in the next 31-60 days, and how early can we know?

ML Formulation

Binary classification. For each customer, the model outputs a probability score. Customers are ranked by risk and prioritised for outreach.

Why Churn?

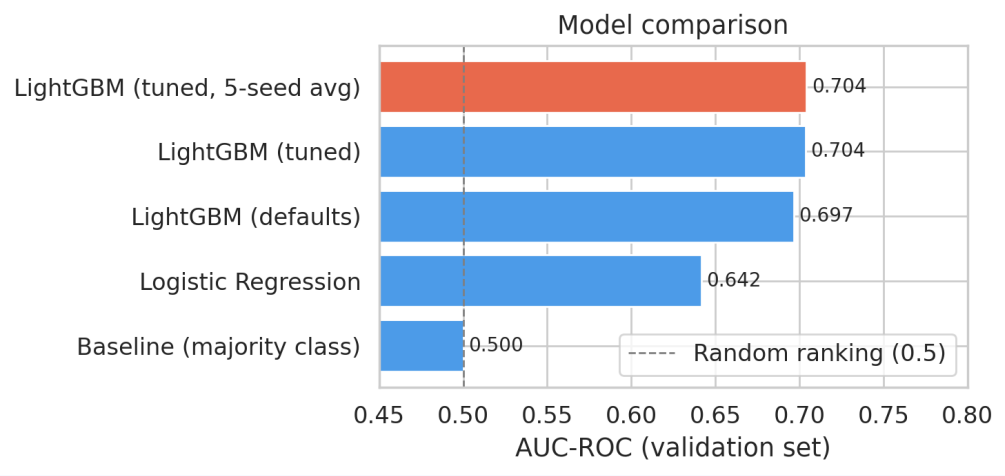
Churn is directly board-relevant and ties to the acquisition-cost case in Slide 3. A 31-60 day prediction window gives enough lead time for retention teams to act before customers are lost.

Evaluation Metric

AUC-ROC (primary) measures ranking quality across all decision thresholds. F1-score (secondary) captures precision-recall balance at the chosen campaign threshold.

We Tested Three Approaches, Then Tuned the Strongest One

Model selection



Baseline

Always predicts the majority class. AUC-ROC = 0.50 by definition. Confirms real signal exists; any useful model must exceed this.

Logistic Regression

Linear model with log-scaled, one-hot inputs; interpretable but limited. Validation AUC 0.642; it cannot capture non-linear effects like the one-year churn spike.

LightGBM (selected)

Gradient boosting; captures non-linear patterns across ~100 correlated features. Tuned with Optuna (5-fold CV, early stopping), 5-seed average: validation AUC 0.704.

Methods: LightGBM (Ke et al., 2017); tuning with Optuna (Akiba et al., 2019); scikit-learn (Pedregosa et al., 2011).

Our LightGBM Model Achieves an AUC-ROC of 0.703: A Real, Moderate Signal Worth Acting On

Test-set results (held-out, never seen during training)

0.703

AUC-ROC

ranking accuracy across all thresholds

~64%

Recall

of churners correctly identified

~64%

Precision

of flagged customers are genuine churners

What does 0.703 AUC-ROC mean?

AUC-ROC ranges 0 to 1: 0.5 is random guessing, 0.7-0.8 is a moderately effective signal, and our 0.703 (95% CI 0.696-0.709) sits at the bottom of that band. The riskiest 10% churn at 3x the average rate, 81.5% in this balanced sample, about 9% at a realistic 3% monthly churn rate.

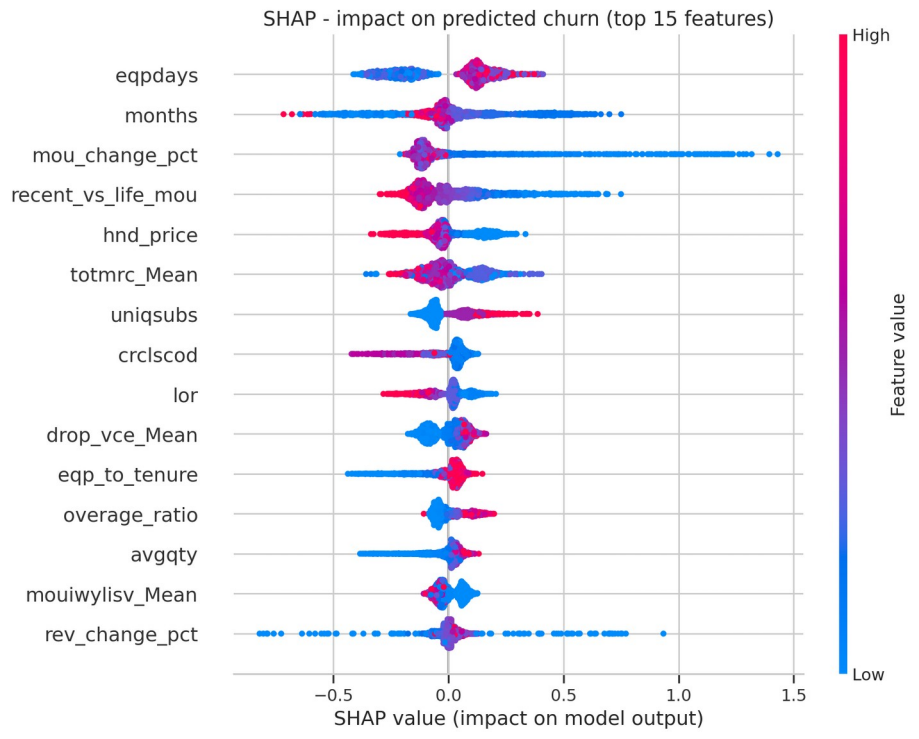
Per 100 actual churners:

~64 correctly flagged
~36 missed (leave with no intervention)

Real-churn-rate figure uses a prior-shift correction (Saerens et al., 2002).

Equipment Age, Tenure, and Usage Trends Drive the Model's Predictions

SHAP beeswarm: impact on predicted churn (top 15 features)



Can act on

Equipment age (**eqpdays**): upgrade offer
Tenure (**months**): pre-renewal offer
Usage trend (**mou_change_pct**): check-in
Plan fit (**totmrc_Mean**, **overage_ratio**)

Informative only

Household subscribers (**uniqsubs**)
Handset price paid (**hnd_price**)
Credit class (**crclscod**)
Length of residence (**lor**)

Handset age and tenure lead, with usage change relative to the customer's own history close behind, matching the EDA. Old handsets (red) push toward churn.

SHAP: Lundberg & Lee (2017).

A Sharply Targeted Campaign Pays Off: Offer Cost Sets the Scale

$23k-$270k

net per 100k customers over 24 months, as offer cost falls from 1 to 1/4 month's revenue

Equipment Upgrade Offer

Target: positive-value customers with handsets 1+ year old
Action: early upgrade, device discount, or bundle upgrade
Rationale: #1 model driver; churn rises from 40% to 58%

Pre-Renewal Loyalty Offer

Target: positive-value customers at 10-15 months tenure
Action: loyalty discount, renewal perk, or plan review
Rationale: churn jumps from 34% to 64% at one year

Assumptions and what they imply

Real churn rate: **3%/month (assumed; sample is 50/50)**
Top-10% precision: **~9% real (81.5% as sampled)**
Retention success / horizon: **30% / 24 months**
Break-even at 1-month offer: **churn prob. > 13.9%**
Top-10% campaign at 1-month offer: **-$0.2M**
Contact if value > 0: **0.7% of base, +$23k**

How it works:



Score monthly and contact only customers whose expected value is positive. Cheaper offers widen that group: at 1/4 month's revenue, ~29% of the base qualifies.

Sources: prior-shift correction (Saerens et al., 2002); expected-value targeting (Elkan, 2001; Verbeke et al., 2012).

This Is a Proof of Concept: Here Is What Strengthens It Before Production

Limitations and recommended next steps

AUC-ROC of 0.703 is moderate

More data (support-ticket text, network-quality logs, competitor pricing) could lift performance. But the business case moves far more with offer cost and retention success than with accuracy.

Key assumptions need validation

Real churn rate (3%/month assumed), retention success (30%) and offer cost decide the case. Replace them with Telecom's figures and pilot results before budgeting.

Run a pilot before full rollout

Randomised pilot: ~490 customers per arm (offer vs. no offer) detects a 30% cut in churn. It measures the real retention rate and cost per save, and gives the data for uplift modelling.

Phased deployment path

Phase 1 (months 1-2): connect the model to live customer records; monthly scoring.
Phase 2 (months 2-4): randomised pilot; measure retention rate and cost per save.
Phase 3 (months 4-6): reset the contact threshold from pilot results; build an uplift model; retrain quarterly.

Uplift modelling: Kunzel et al. (2019).

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